Who Is Imitating Whom? Reframing Bidirectional Alignment as Social Learning and Cultural Transmission

Sachit Mahajan
Computational Social Science, ETH Zurich
Zurich, Switzerland
sachit.mahajan@gess.ethz.ch

Abstract

The alignment of AI systems with human values is increasingly treated as an optimization problem: specify the right reward signal, train with the right feedback, and the system will converge on human preferences. This paper argues that such a framing fundamentally mischaracterizes the dynamics at play when millions of people interact with AI systems daily. Drawing on social learning theory, cultural evolution research, and the diffusion of innovations literature, this position paper reframes bidirectional human-AI alignment as a process of *cultural transmission*—one in which both humans and AI systems simultaneously shape and reshape each other's cultural patterns. This reframing surfaces three critical blind spots in current alignment research: (1) AI systems perform a functionally analogous form of cultural reproduction that strips away the contextual intelligence that makes human imitation culturally adaptive; (2) the fluency and authority of AI outputs may degrade humans' capacity for selective, critical engagement—shifting them from innovative evaluation to high-fidelity copying; and (3) the largely private, dyadic nature of most human-AI interaction lacks the structured social mediation that has historically governed cultural diffusion. The paper contributes a theoretical framework, four design principles, and a research agenda aimed at restoring the conditions under which human-AI mutual learning can be genuinely intelligent and culturally beneficial.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.

Keywords

Human-AI alignment, social learning, imitation, value-sensitive design, human-AI interaction, participatory design

ACM Reference Format:
Sachit Mahajan. 2026. Who Is Imitating Whom? Reframing Bidirectional Alignment as Social Learning and Cultural Transmission. In *Proceedings of Bidirectional Human-AI Alignment Workshop at ACM CHI Conference on Human Factors in Computing Systems (CHI '26 BiAlign Workshop).* ACM, New York, NY, USA, 6 pages. https://doi.org/XXXXXXX.XXXXXXX

CHI '26 BiAlign Workshop, Barcelona, Spain

ACM ISBN 978-1-4503-XXXX-X/2018/06
https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

The dominant paradigm in AI alignment—most clearly seen in approaches like reinforcement learning from human feedback (RLHF) [9, 23] and constitutional AI [3]—treats alignment as a matter of getting the training signal right. Specify what humans want, provide enough feedback, and the model will learn to behave accordingly. At its core, this is an engineering framing: alignment is cast as an optimization target, and the central challenge is technical. This paper focuses specifically on alignment with shared cultural values and collective norms, rather than on alignment with individual user preferences. It is at this collective level that cultural transmission theory becomes most directly relevant.

Approaches like RLHF and constitutional AI have yielded real progress. Yet they rest on a limited understanding of what actually occurs when millions of people interact with AI systems on a daily basis. What unfolds in these interactions is not merely the "use" of a tool that has been aligned or misaligned with human preferences. Rather, it constitutes a process of *mutual cultural transmission*—a bidirectional dynamic in which AI systems absorb and reproduce human cultural patterns, while humans in turn absorb and reproduce AI-generated patterns, each reshaping the other in ways that remain difficult to observe and poorly understood.

This claim draws its force from established theoretical traditions. Social learning theory [5], cultural evolution research [13, 26], and the diffusion of innovations literature [27] offer precise, empirically grounded frameworks for understanding how behaviors, values, and knowledge propagate between agents. These frameworks have been developed and refined over decades across the behavioral sciences, yet they remain almost entirely absent from the alignment literature. The result is a field that lacks the conceptual vocabulary to diagnose some of its most consequential problems—particularly those that fall squarely within the domain of human-computer interaction (HCI).

This position paper advances three arguments. First, that treating alignment as cultural transmission reveals blind spots that current technical and philosophical framings overlook. Second, that HCI is uniquely positioned to address these blind spots, given the discipline's sustained engagement with situated interaction [32], reflective design [30], value-sensitive design [11], and participatory methods [22]. Third, that concrete design principles can be derived from social learning theory to support genuinely bidirectional alignment. The paper's contributions are:

(1) A **theoretical reframing** of bidirectional human-AI alignment as a cultural transmission process, grounded in social learning theory.

(2) The identification of **three blind spots** in current alignment research—concerning evaluation, human cognition, and social infrastructure—that this reframing reveals.
(3) **Four design principles** for human-AI interaction that support intelligent, selective, and socially embedded cultural exchange.
(4) A **research agenda** connecting HCI methods to the study of alignment as a lived cultural practice.

2 Background: Imitation as Intelligent Cultural Practice

The insight that imitation requires intelligence—that it is not passive mimicry but a cognitively demanding, socially embedded act—has deep roots in the behavioral and cognitive sciences. Bandura's foundational work on social learning [4, 5] established that observational learning involves at minimum four distinct processes: attention to the model, retention of observed behavior, motor reproduction, and motivation. Crucially, the learner does not copy everything. Imitation is selective: the observer decides what is worth reproducing based on context, perceived consequences, and the status and competence of the model.

Subsequent work in cultural evolution deepened this picture considerably. Tomasello [34] argued that humans' unique capacity for cumulative culture depends precisely on the ability to imitate *with understanding*—to grasp the goals and intentions behind observed actions, not merely their surface form. Building on this foundation, Legare and Nielsen [18] articulated what they term the "dual engines" of cultural learning: *high-fidelity imitation*, which preserves conventional and ritualistic knowledge across generations, and *innovation*, which adapts and improves instrumental knowledge through creative modification. These are not simply ends of a single continuum; they represent functionally distinct cognitive strategies deployed in response to different contextual cues. Legare [17] further demonstrated that even young children toggle between these modes with remarkable sophistication, depending on whether the context signals a social convention (copy faithfully) or a causal problem (experiment and modify).

This distinction carries significant weight for alignment. High-fidelity imitation serves cultural cohesion; innovation serves adaptation and progress. A healthy cultural learning ecology requires both, deployed in the right contexts. When the balance is disrupted—when people copy faithfully where they ought to innovate, or innovate where they ought to preserve—cultural systems become either brittle or chaotic [13]. Yiu et al. [35] have recently applied this very framework to large language models, arguing that such systems function as powerful "imitation engines" that can reproduce existing cultural knowledge but lack the capacity for genuine innovation—the second engine that children deploy naturally.

The diffusion of innovations literature adds a further critical dimension. Ryan and Gross's [28] classic study of hybrid seed corn adoption among Iowa farmers demonstrated that even the uptake of clearly superior innovations is not automatic but proceeds through social networks, mediated by peer observation, trust, and staged decision-making. Rogers [27] formalized this into a five-stage adoption process—knowledge, persuasion, decision, implementation, confirmation—that unfolds within social systems with their own norms, opinion leaders, and communication channels. The central insight is that cultural adoption is not a dyadic process between an individual and an innovation; it is fundamentally *social*, governed by community deliberation and collective sense-making. Crucially, Rogers' framework assumes a conscious adopter encountering a discrete, identifiable innovation—conditions that, as the next section argues, do not straightforwardly hold for human-AI interaction.

For HCI, these traditions raise a pointed question: if interaction design shapes how people encounter, evaluate, and adopt new practices, then the design of human-AI interaction directly structures the cultural transmission process through which alignment is—or fails to be—achieved.

3 AI as a Participant in Cultural Transmission

With this theoretical background in place, consider what modern AI systems actually represent in cultural terms. Large language models are trained on vast corpora of human-generated text—the accumulated output of human culture [6]. Through this training, they absorb statistical patterns of language use, reasoning, aesthetic preference, moral framing, and social convention. They then reproduce these patterns in interaction with individual users, who in turn incorporate AI-generated language, reasoning, and framings into their own thought and practice.

3.1 A Necessary Clarification: Functional Analogy, Not Literal Equivalence

Before proceeding, a clarification is essential. To describe AI systems as participating in cultural transmission is to make a claim about *functional consequences*, not about underlying cognitive mechanisms. AI systems do not "imitate" in Bandura's sense: they lack intentionality, goal-directedness, and the capacity to understand the model as an agent. Statistical pattern learning is mechanistically distinct from observational learning. However, the *cultural-level effects* are functionally analogous. When an AI system reproduces linguistic patterns, value framings, and reasoning structures drawn from human-generated data, and when humans adopt these reproductions into their own practice, the result is a transmission of cultural patterns between agents—regardless of whether the AI "understands" what it is transmitting. This is precisely the point Bender et al. [6] raised with the "stochastic parrots" metaphor: systems that produce form without meaning. The social learning framework offered here provides a more specific vocabulary for analyzing *what kind* of cultural reproduction this constitutes and *what consequences* it carries—moving beyond the binary of "meaningful understanding" versus "mere parroting" to examine the specific dimensions along which AI cultural reproduction falls short and the specific ways it affects human recipients.

With this clarification in place, three problems emerge.

3.2 The Shallow Reproduction Problem

When AI systems learn from human data, they perform what might be termed *aggregate surface reproduction*. The "model" being reproduced is not any particular person, community, or coherent value system—it is a statistical composite of millions of voices, flattened into distributional patterns [6]. Bandura's framework [5] would predict that the quality of observational learning depends on the coherence and observability of the model. But the cultural source that AI reproduces has no coherent intentions, no situated context, no social identity. In cultural evolutionary terms, it is a blurred signal—a cultural average that may not represent any actual community's values or practices [12].

RLHF and related techniques [23] attempt to sharpen this signal through human feedback. However, the feedback itself is typically provided by paid annotators operating under time pressure, with limited context about the downstream uses of the system [8]. This bears little resemblance to the rich, situated, socially embedded feedback that governs cultural transmission in human communities. To borrow Suchman's [32] framing, it is feedback stripped of the situated actions that give it meaning.

The result is cultural reproduction that produces outputs which *appear* culturally fluent—grammatically correct, contextually plausible, stylistically appropriate—but which lack the deeper understanding of goals, contexts, and consequences [19]. As Yiu et al. [35] argue, the system has mastered one engine of cultural learning—high-fidelity reproduction—while entirely lacking the other: the capacity for genuine innovation grounded in causal understanding. This matters for alignment because an AI system that reproduces cultural patterns without understanding their contextual appropriateness will inevitably transmit those patterns into contexts where they do not belong.

3.3 The Uncritical Adoption Problem

On the human side, the picture is equally concerning. When people interact with AI systems, they encounter outputs that are fluent, confident, and often impressively competent. The conditions are, in Bandura's [5] terms, precisely those that promote uncritical modeling: a high-status model (AI is increasingly perceived as authoritative [33]), demonstrating apparently successful behavior (the outputs sound right, and the task gets done faster), with clear reinforcement (efficiency gains are immediate and tangible).

Research on automation bias [24] has long documented the human tendency to defer to automated systems, even when those systems make mistakes. The social learning perspective, however, identifies a qualitatively distinct concern. Automation bias is primarily an *epistemic* problem: a miscalibration of trust leading to excessive reliance. The concern raised here is *procedural*: that AI's authoritative fluency may shift the very cognitive strategy users deploy, from the evaluative, causally-oriented mode that Legare and Nielsen [18] associate with innovation to the faithful, convention-preserving mode they associate with high-fidelity imitation. The difference matters because trust miscalibration can be addressed by improving calibration (e.g., displaying confidence scores), whereas a strategy shift requires restructuring the interaction itself to re-engage evaluative cognition.

An important caveat is necessary here. Legare and Nielsen's framework was developed through research on children's cultural learning, primarily in the age range of two to seven years. Adults possess metacognitive capacities and social learning strategies that children do not, which may buffer against uncritical adoption. Extending the dual-engines framework to adult-AI interaction is therefore a *theoretical proposition* that requires empirical validation—one that the research agenda below is designed to address. Nevertheless, empirical evidence already suggests that the direction of the effect holds for adults: interaction with opinionated language models can shift users' expressed views [14], and co-writing with AI systems measurably alters the linguistic and conceptual patterns people produce [15]. The broader concern is that AI fluency creates what might be termed an *imitation trap*: the outputs are good enough to adopt wholesale, removing the friction that normally triggers selective evaluation and adaptation. Sengers et al.'s [30] call for *reflective design*—design that supports critical awareness of one's own assumptions and practices—takes on new urgency in this context.

3.4 The Erosion of Social Mediation

Rogers' [27] diffusion framework highlights a third problem that is rarely foregrounded in alignment discourse: in many domains, the adoption of new practices is mediated by social networks. People observe peers, consult trusted advisors, witness consequences in others' lives, and make adoption decisions within a web of social accountability. Ryan and Gross [28] found that even Iowa farmers—a relatively homogeneous population adopting a clearly beneficial innovation—took years to adopt hybrid corn, proceeding through stages of awareness, trial, and community validation. Crucially, Rogers' model presumes an adopter who recognizes an adoption decision, encounters a discrete and identifiable innovation, and can observe outcomes in public or shared settings—conditions under which staged social mediation can operate.

The analogy to innovation diffusion breaks down in ways that are themselves revealing. AI-mediated cultural uptake is continuous rather than discrete, often non-deliberative rather than reflective, and largely invisible to peers. There are no trial plots to inspect, no harvest yields to compare. The very conditions that Rogers identified as enabling staged diffusion—time for deliberation, observable consequences, peer validation—are structurally weakened in many human–AI interactions. While some informal social mediation does occur—teams discuss AI-generated code, colleagues share outputs on workplace channels, educators debate AI use in classrooms—these processes are *unstructured and incidental*, falling far short of the deliberative community dynamics Rogers described. The core of the interaction—a person typing a prompt and receiving a response—remains a private, dyadic exchange.

The concern, then, is not that AI-shaped patterns are diffusing too slowly through social networks, but that they can bypass the logic of diffusion entirely: propagating at the tempo of chat interaction and at a reach enabled by replication and platform scale, without the social scaffolding that has historically made adoption processes corrigible and accountable.

Boyd and Richerson [7] argued that cultural evolution depends on a balance between individual learning (costly but accurate) and social learning (cheap but dependent on the quality of available models). AI interaction disrupts this balance in an unprecedented way: it offers what *feels* like social learning—engaging with an apparently knowledgeable agent—but without the structured social infrastructure that normally ensures quality control over transmitted culture. The participatory design tradition in HCI [22, 29] has long advocated for community involvement in shaping the technologies that affect collective life; the absence of designed social infrastructure for AI-mediated cultural exchange represents a direct challenge to this principle.

4 From Problems to Blind Spots: What Current Alignment Misses

The problems described above are not merely interesting observations; they point to specific gaps in how the alignment field conceptualizes and evaluates its own success. This section articulates what current approaches fail to measure, theorize, or design for.

Evaluation gap: alignment metrics do not capture cultural adaptiveness. Current alignment evaluation—benchmarks, red-teaming, human preference ratings [23]—assesses whether AI outputs match human expectations at the surface level. Such evaluation does not assess whether the system has learned the *contextual, causal, and intentional* structure that makes human cultural knowledge adaptive. A system can score perfectly on alignment benchmarks while performing what is, in cultural evolutionary terms, shallow and brittle reproduction—transmitting cultural patterns into contexts where they are inappropriate. The field lacks evaluation frameworks that ask Legare's [17] question: Does the system know *when* to reproduce convention faithfully and when to adapt to situated context? Developing such frameworks—which would require testing AI behavior across varied contextual conditions, not just aggregate preference matching—represents a concrete research opportunity for the HCI community.

Cognitive gap: the human side of alignment is theorized normatively, not mechanistically. The "aligning humans with AI" direction articulated in the BiAlign framework is a welcome corrective, but even this framing tends to preserve *agency* in abstract, normative terms [2, 31]. What remains missing is attention to the specific *cognitive mechanisms* through which people evaluate, adapt, or uncritically absorb AI outputs—the attentional, evaluative, and motivational processes that Bandura [5] identified as constitutive of observational learning. Designing for "human agency" requires understanding what cognitive capacities that agency depends on, and whether those capacities are being exercised or atrophied through sustained human-AI interaction. HCI research on reflective design [30] and critical technical practice [1] offers relevant precedent, but these threads have not been connected to the alignment discourse.

Infrastructure gap: there is no designed social layer for AI cultural transmission. The fact that AI-mediated cultural change lacks structured social mediation is not just a descriptive observation; it is a *design absence* with consequences. Floridi et al. [10] and others have called for societal-level alignment, and Rahwan et al. [25] have proposed studying "machine behaviour" as a social phenomenon. But neither has proposed *design interventions* that would re-introduce the kind of structured community deliberation that Rogers [27] identified as essential for healthy cultural diffusion. Value-sensitive design [11] has long distinguished between direct stakeholders, indirect stakeholders, and broader societal impacts; applying this tripartite analysis to the cultural transmission dynamics of human-AI interaction could yield concrete design requirements for the missing social infrastructure. It is worth noting that some current alignment approaches, such as constitutional AI [3], do involve carefully authored value guidelines during training. The gap identified here is not about that stage, but about what happens afterward: how AI-shaped patterns spread through millions of daily interactions where no such deliberation is present.

5 A Social Learning Framework for Bidirectional Alignment

This section proposes a *social learning framework* that takes the cultural transmission dynamics described above seriously. The framework is intended to complement, not replace, existing technical approaches, but to reorient research attention toward questions that current paradigms neglect—questions that fall naturally within HCI's disciplinary strengths.

5.1 Design Principles

Drawing on the theoretical resources outlined above, four design principles are proposed for human-AI interaction systems that aspire to genuinely bidirectional alignment.

Principle 1: Design for innovation, not just consumption. Interfaces should be structured to trigger the evaluative, causally-oriented reasoning that Legare and Nielsen [18] associate with innovation—the second engine of cultural learning. Concretely, consider a writing assistant that, rather than presenting a single polished draft, presents three alternative framings of the same content and asks the user: "Which framing best fits your audience and purpose? Why?" This small interaction shift changes the cognitive task from acceptance/rejection (which defaults toward high-fidelity copying of the most fluent option) to comparative evaluation (which engages causal reasoning about context and purpose). The goal is not to add friction for its own sake, but to restore the conditions under which engagement with AI outputs is intelligent rather than reflexive. This principle extends Sengers et al.'s [30] reflective design agenda into the specific domain of AI alignment.

Principle 2: Make the cultural provenance of AI outputs visible. If AI systems perform aggregate reproduction of human culture, users should have some visibility into *what cultural patterns* are being reproduced. This is distinct from conventional explainability, which focuses on model reasoning or feature attribution. Cultural provenance would involve indicating, for example, that a particular framing reflects dominant patterns in a specific professional community, or that a recommendation is grounded in values prevalent in certain cultural contexts but not others. This draws on the idea of provenance [20], value-sensitive design's [11] emphasis on surfacing embedded values and connects to broader work on AI transparency [2].

Principle 3: Re-embed AI interaction in social networks. If Rogers [27] is right that healthy cultural diffusion requires social mediation, then the absence of structured social infrastructure for human-AI cultural exchange is itself a design problem. Interaction systems could be designed to reconnect individual AI interactions to community deliberation—for example, by enabling teams or communities to collectively evaluate and curate AI-generated outputs, or by making visible how peers have responded to similar AI suggestions. The participatory design tradition [22, 29] offers extensive methodological resources for structuring such collective engagement. Lee et al.'s [16] WeBuildAI framework, which enables communities to collectively define algorithmic governance criteria, provides a concrete precedent for what participatory alignment processes might look like. This principle reframes alignment not as a property of the AI system alone, but as a *community practice* that requires social infrastructure.

Principle 4: Monitor alignment drift, not just alignment state. Most alignment evaluation is static: does the system meet specifications at a given point in time? A cultural transmission framework highlights the need for *longitudinal* monitoring of how both AI behavior and human behavior co-evolve through interaction. Drawing on Mesoudi's [21] methods for studying cultural evolution, researchers could track how AI-generated patterns diffuse through user populations over time, and how users' own linguistic, reasoning, and evaluative practices shift through sustained AI interaction. This principle is essential for addressing the dynamic co-evolution that the BiAlign framework identifies as a core concern, and calls for longitudinal HCI research methodologies that extend beyond single-session laboratory studies.

5.2 A Research Agenda

The social learning framework opens several research directions that remain underexplored and that align with HCI's methodological strengths.

Empirical studies of cognitive strategy in human-AI interaction. Do users engage in high-fidelity adoption or innovative adaptation of AI outputs, and what interaction design factors modulate this? Legare's [17] developmental methodology—presenting participants with tasks that require either faithful reproduction or adaptive modification—could be adapted for adult-AI contexts, with appropriate attention to the metacognitive differences between child and adult learners. Think-aloud protocols and interaction log analysis could reveal when and why users shift between cognitive strategies, and whether specific interface features promote one mode over the other.

Cultural diffusion mapping of AI-generated patterns. How do AI-generated linguistic forms, reasoning strategies, or value framings propagate through communities of users? Existing cultural evolution methods [13, 21] for tracking the transmission of cultural variants could be applied at scale using computational approaches, complemented by qualitative fieldwork in communities where AI use is intensive.

Participatory alignment processes that restore social mediation. Rather than relying on expert annotators or automated methods, alignment evaluation could involve structured community deliberation about what "good" AI behavior looks like in specific contexts [16, 22]. Such processes would draw on HCI's extensive experience with participatory and co-design methods, and would operationalize the principle that alignment is a community practice, not merely a technical specification.

Longitudinal studies of cognitive and cultural effects. How does sustained interaction with AI systems affect users' capacity for critical evaluation, creative adaptation, and independent reasoning over weeks and months? This calls for the kind of ecologically valid, long-duration research designs that Suchman [32] has long advocated for studying human-machine interaction, and that diary studies and experience sampling methods can support.

6 Discussion

An important clarification is in order regarding the scope of this argument. The position advanced here does not claim that AI alignment research should abandon its technical methods. RLHF, constitutional AI, and related approaches address real and important problems. Nor does social learning theory provide a complete alternative framework for alignment. Rather, the claim is that the field has a significant theoretical blind spot. By treating alignment primarily as an optimization problem, it has neglected the cultural dynamics through which AI systems and human users are actually shaping each other. This paper is not the first to observe that alignment resists reduction to optimization. Gabriel [12] has articulated the philosophical difficulty of specifying human values, and Casper et al. [8] have catalogued fundamental limitations of RLHF. The contribution here is not that observation itself, but the identification of a specific theoretical tradition, social learning and cultural evolution, that provides analytical tools for understanding the bidirectional dynamics these prior critiques leave unaddressed.

The social learning tradition—from Bandura's [5] analysis of observational learning, through Legare's [17, 18] work on the dual engines of cultural learning, to Rogers' [27] diffusion framework—offers conceptual tools that the alignment field urgently needs. Equally, HCI's traditions of reflective design [30], value-sensitive design [11], and participatory methods [22] offer *design* tools for translating these theoretical insights into actionable interventions.

The practical stakes are considerable. If AI interaction is indeed shifting the cognitive strategies through which people engage with cultural inputs—from evaluative innovation toward high-fidelity reproduction—then even technically well-aligned AI systems could produce negative cultural outcomes at scale. Conversely, if interaction systems can be designed to support selective, critical, socially embedded engagement with AI outputs, the bidirectional alignment process could become genuinely productive—not just safe, but culturally enriching.

Two limitations of this analysis deserve acknowledgment. First, the theoretical framework advanced here is, at this stage, precisely that—a theoretical framework. The empirical claims about cognitive strategy shifting, and cultural diffusion effects are supported by converging evidence from adjacent fields [14, 15, 24], but they have not been directly tested in the context of human-AI alignment. The research agenda outlined above is designed to address this gap. Second, the extension of Legare and Nielsen's developmental framework to adult-AI interaction is a theoretical proposition whose boundary conditions remain to be established empirically. Adults bring metacognitive resources that may moderate the effects described here, and identifying when and for whom these effects are strongest is itself a critical research question.

7 Conclusion

Alignment is not just an engineering problem. It is a cultural process—one that involves mutual shaping between humans and AI systems, operating at unprecedented scale, speed, and intimacy. Drawing on social learning theory and cultural evolution research, this paper has identified three gaps in current alignment thinking—in evaluation, in attention to human cognition, and in social infrastructure—and has proposed a framework, design principles, and research agenda to address them. The most urgent question this analysis raises is not whether AI systems are aligned with human values in the abstract, but whether the interaction patterns through which alignment is enacted support the kind of intelligent, selective, socially accountable cultural exchange on which human flourishing has always depended. That question belongs to HCI, and the BiAlign community is well positioned to take it up.

References

[1] Philip Agre. 1997. *Computation and human experience.* Cambridge University Press.
[2] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, et al. 2019. Guidelines for human-AI interaction. In *Proceedings of the 2019 chi conference on human factors in computing systems.* 1–13.
[3] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. 2022. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073* (2022).
[4] Albert Bandura. 1965. Influence of models' reinforcement contingencies on the acquisition of imitative responses. *Journal of personality and social psychology* 1, 6 (1965), 589.
[5] Albert Bandura and Richard H Walters. 1977. *Social learning theory.* Vol. 1. Prentice hall Englewood Cliffs, NJ.
[6] Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big?. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency.* 610–623.
[7] Robert Boyd and Peter J Richerson. 1988. *Culture and the evolutionary process.* University of Chicago press.
[8] Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, Jérémy Scheurer, Javier Rando, Rachel Freedman, Tomasz Korbak, David Lindner, Pedro Freire, et al. 2023. Open problems and fundamental limitations of reinforcement learning from human feedback. *arXiv preprint arXiv:2307.15217* (2023).
[9] Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. *Advances in neural information processing systems* 30 (2017).
[10] Luciano Floridi, Josh Cowls, Monica Beltrametti, Raja Chatila, Patrice Chazerand, Virginia Dignum, Christoph Luetge, Robert Madelin, Ugo Pagallo, Francesca Rossi, et al. 2018. AI4People—An ethical framework for a good AI society: Opportunities, risks, principles, and recommendations. *Minds and machines* 28, 4 (2018), 689–707.
[11] Batya Friedman and David G Hendry. 2019. *Value sensitive design: Shaping technology with moral imagination.* Mit Press.
[12] Iason Gabriel. 2020. Artificial intelligence, values, and alignment. *Minds and machines* 30, 3 (2020), 411–437.
[13] Joseph Henrich. 2015. The secret of our success: How culture is driving human evolution, domesticating our species, and making us smarter. In *The secret of our success.* princeton University press.
[14] Maurice Jakesch, Advait Bhat, Daniel Buschek, Lior Zalmanson, and Mor Naaman. 2023. Co-writing with opinionated language models affects users' views. In *Proceedings of the 2023 CHI conference on human factors in computing systems.* 1–15.
[15] Mina Lee, Katy Ilonka Gero, John Joon Young Chung, Simon Buckingham Shum, Vipul Raheja, Hua Shen, Subhashini Venugopalan, Thiemo Wambsganss, David Zhou, Emad A Alghamdi, et al. 2024. A design space for intelligent and interactive writing assistants. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems.* 1–35.
[16] Min Kyung Lee, Daniel Kusbit, Anson Kahng, Ji Tae Kim, Xinran Yuan, Allissa Chan, Daniel See, Ritesh Noothigattu, Siheon Lee, Alexandros Psomas, et al. 2019. WeBuildAI: Participatory framework for algorithmic governance. *Proceedings of the ACM on human-computer interaction* 3, CSCW (2019), 1–35.
[17] Cristine H Legare. 2017. Cumulative cultural learning: Development and diversity. *Proceedings of the National Academy of Sciences* 114, 30 (2017), 7877–7883.
[18] Cristine H Legare and Mark Nielsen. 2015. Imitation and innovation: The dual engines of cultural learning. *Trends in cognitive sciences* 19, 11 (2015), 688–699.
[19] Sachit Mahajan. 2025. The democratization dilemma: When everyone is an expert, who do we trust? *Humanities and Social Sciences Communications* 12, 1 (2025), 1–5.
[20] Sachit Mahajan. 2025. Participatory provenance: Measuring input transformation in ai-mediated collective synthesis. *Available at SSRN 5709064* (2025).
[21] Alex Mesoudi. 2011. Cultural evolution: How Darwinian theory can explain human culture and synthesize the social sciences. In *Cultural Evolution.* University of Chicago Press.
[22] Michael J Muller and Sarah Kuhn. 1993. Participatory design. *Commun. ACM* 36, 6 (1993), 24–28.
[23] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. *Advances in neural information processing systems* 35 (2022), 27730–27744.
[24] Raja Parasuraman and Dietrich H Manzey. 2010. Complacency and bias in human use of automation: An attentional integration. *Human factors* 52, 3 (2010), 381–410.
[25] Iyad Rahwan, Manuel Cebrian, Nick Obradovich, Josh Bongard, Jean-François Bonnefon, Cynthia Breazeal, Jacob W Crandall, Nicholas A Christakis, Iain D Couzin, Matthew O Jackson, et al. 2019. Machine behaviour. *Nature* 568, 7753 (2019), 477–486.
[26] Peter J Richerson and Robert Boyd. 2008. *Not by genes alone: How culture transformed human evolution.* University of Chicago press.
[27] Everett M Rogers, Arvind Singhal, and Margaret M Quinlan. 2014. Diffusion of innovations. In *An integrated approach to communication theory and research.* Routledge, 432–448.
[28] Bryce Ryan and Neal C Gross. 1943. The diffusion of hybrid seed corn in two Iowa communities. *Rural sociology* 8, 1 (1943), 15.
[29] Douglas Schuler and Aki Namioka. 1993. *Participatory design: Principles and practices.* CRC press.
[30] Phoebe Sengers, Kirsten Boehner, Shay David, and Joseph'Jofish' Kaye. 2005. Reflective design. In *Proceedings of the 4th decennial conference on Critical computing: between sense and sensibility.* 49–58.
[31] Ben Shneiderman. 2020. Human-centered artificial intelligence: Reliable, safe & trustworthy. *International Journal of Human–Computer Interaction* 36, 6 (2020), 495–504.
[32] Lucille Alice Suchman. 2007. *Human-machine reconfigurations: Plans and situated actions.* Cambridge university press.
[33] Cass R Sunstein. 2019. Conformity: The power of social influences. In *Conformity.* New York University Press.
[34] Michael Tomasello. 2009. *The cultural origins of human cognition.* Harvard university press.
[35] Eunice Yiu, Eliza Kosoy, and Alison Gopnik. 2024. Transmission versus truth, imitation versus innovation: What children can do that large language and language-and-vision models cannot (yet). *Perspectives on Psychological Science* 19, 5 (2024), 874–883.